Jadon Au-Yeung

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Skills & Certifications

Programming Languages: Python, JavaScript, TypeScript, C++, Rust, R, HTML, CSS, SQL

Python Stacks: Pandas, NumPy, Matplotlib, Scikit-learn, FastAPI, Pydantic, SQLAlchemy, OAuth2, LangChain

Tools & Data: Jupyter Notebook, Git, VS Code, PGAdmin4, MongoDB Compass, Databricks

Certifications: Machine Learning with Python (FreeCodeCamp); Associate AI Engineer (DataCamp)

Skills: Prompt Engineering, Agentic AI, Problem Solving, Data Analysis, Database Management, Cloud Computing

Projects

Stride - Mobile App — **Apr 2026 – Present**

- Designed and deployed a RESTful API using FastAPI and PostgreSQL, implementing a full authentication system with JWT (HS256) token issuance and verification, Argon2 password hashing, and role-based route protection via reusable dependency injection — securing user, steps, and product endpoints across the application.
- Integrated Stripe's payment processing pipeline end-to-end, building a checkout session creation endpoint and a webhook handler that validates Stripe signatures and persists completed payment records to a relational database with full event-driven data integrity.
- Architected a normalised PostgreSQL schema using SQLAlchemy ORM with cascading foreign key relationships across Users, stepsHistory, Payment, and Products tables, paired with Pydantic schema validation for all inbound and outbound data, and environment-based configuration management via pydantic-settings for secure credential handling across environments.

Why do the Sacramento Kings Suck? — **Apr 2026 – Apr 2026**

- Developed an LSTM autoencoder in Keras/TensorFlow to encode 66-game regular season sequences into dense 16-dimensional latent representations, enabling unsupervised comparison of team seasons across different eras
- Applied KMeans clustering and PCA dimensionality reduction to surface structural similarities between Sacramento Kings seasons and championship-winning teams since 2000, identifying that Kings seasons cluster as a statistically distinct group from champions across offensive rating, pace, and win patterns
- Built a RAG agent using LangChain, ChromaDB, and Claude (Anthropic) to answer natural language queries over 50+ years of NBA transaction records — implementing RAG Fusion with reciprocal rank fusion to broaden semantic retrieval coverage, and enforcing structured JSON output via Pydantic schemas

UFC Parlay Predictor — **Apr 2026 - May 2026**

- Scraped and cleaned 10,000+ UFC fight statistics using Pandas to build a structured dataset, engineering features such as fighter matchup history and using boolean masks to detect rematches.
- Designed a multi-LLM pipeline to ingest and parse raw UFC parlay text, using prompt engineering and structured JSON outputs to reliably extract fighters, odds, and bet types for downstream modeling.
- Implemented output validation and correction layers for LLM responses, enforcing schema constraints and reducing extraction errors through rule-based checks and fallback parsing strategies.
- Trained multiple XGBoost models on 5,000+ UFC fights to predict outcomes (win/loss, over/under, and survival time), integrating LLM-extracted features into a unified predictive betting framework.

Canadian Athlete Network — **Feb 2026 – Apr 2026**

- Engineered a high-performance REST API with FastAPI, supporting advanced athlete discovery through K-Nearest Neighbours (KNN) search and secure email-based authentication.
- Containerized the full application stack by separating frontend and backend services into independent Dockerfiles and orchestrating them with Docker Compose for reliable, reproducible deployments.

Professional Experience

Yard Associate Team Member — **Apr 2025 – Aug 2025**

- Worked with a team of students to complete 50+ customer orders daily in a fast-paced environment while maintaining accuracy and safety.
- Loaded and organized heavy hardware material (lumber, concrete, bricks, etc.) to support efficient customer pickup, while conducting daily cleanups to comply with safety and operational standards.
- Applied supervisor feedback to effectively improve workflow and productivity, building strong team cohesion through clear communication and collaborative problem-solving.

Education

York University - Schulich School of Business — *Graduating 2028*

Bachelor of Science – Data Science, Business Stream (HBSc) — Toronto, Ontario